

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2024 P1HR Worked Solutions

May 2024 | P1HR

27 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Statistics Toolkit**

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1 Here are six cards.

Five of the cards have a number written on them.

16	15	3	2	9	
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Work out the number that should be written on the last card so that the mean of the six numbers will be 11

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Statistics Toolkit**

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WORKED SOLUTION

Dr. Eslam Ahmed

1. The total of the 6

The total of the 6 cards is

$$6 \times 11 = 66$$

2. The known total is

The known total is

$$16 + 15 + 3 + 2 + 9 = 45$$

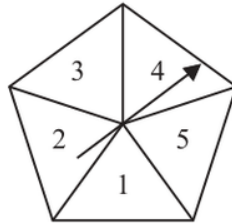
$$x = 66 - 45 = 21$$

FINAL ANSWER

$$x = 21.$$

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

2 Here is a biased spinner.

The table gives information about the probability that, when the spinner is spun once, it will land on each number.

Number	1	2	3	4	5
Probability	$2x$	0.27	0.04	x	0.12

Alexis is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on an odd number.

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 4****TOPIC: Probability Toolkit**

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WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the probability of landing**

Let the probability of landing on 4 be x . Then the probability of landing on 1 is $2x$.

$$2x + 0.27 + 0.04 + x + 0.12 = 1$$

$$3x = 0.57$$

$$x = 0.19$$

$$P(\text{odd}) = 0.38 + 0.04 + 0.12 = 0.54$$

2. Expected odd numbers

Expected odd numbers:

$$400 \times 0.54 = 216$$

FINAL ANSWER

0.19, 0.54, 216.

PAST PAPER QUESTION**Q#: 3****Marks: 5****TOPIC: Percentages**

May 2024 P1HR - May 2024 | P1HR

- 3** Norberto sells white loaves of bread and brown loaves of bread.

He sells a total of 200 loaves such that

the number of white loaves sold : the number of brown loaves sold = 3 : 2

Norberto sells the white loaves for £1.50 each.

He sells the brown loaves for £1.75 each.

40% of the price of a white loaf is profit.

60% of the price of a brown loaf is profit.

Work out Norberto's total profit when he sells all 200 loaves.

£.....

(Total for Question 3 is 5 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 5****TOPIC: Percentages**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The ratio parts are**

The ratio parts are

$$3 + 2 = 5$$

2. White loaves

White loaves:

$$200 \times \frac{3}{5} = 120$$

3. Brown loaves

Brown loaves:

$$200 \times \frac{2}{5} = 80$$

4. Profit on each white loaf

Profit on each white loaf:

$$40\% \times 1.50 = 0.60$$

5. Profit on each brown loaf

Profit on each brown loaf:

$$60\% \times 1.75 = 1.05$$

6. Total profit

Total profit:

$$120 \times 0.60 + 80 \times 1.05 = 72 + 84 = 156$$

FINAL ANSWER

156 pounds

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Algebraic Proof**

May 2024 P1HR - May 2024 | P1HR

4 Show that $2\frac{1}{3} \div 5\frac{1}{4} = \frac{4}{9}$

(Total for Question 4 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 4****Marks: 3****TOPIC: Algebraic Proof**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$2\frac{1}{3} = \frac{7}{3}, \quad 5\frac{1}{4} = \frac{21}{4}$$

$$2\frac{1}{3} \div 5\frac{1}{4} = \frac{7}{3} \div \frac{21}{4}$$

$$= \frac{7}{3} \times \frac{4}{21} = \frac{28}{63} = \frac{4}{9}$$

FINAL ANSWER

$$\frac{4}{9}$$

EA

PAST PAPER QUESTION**Q#: 5****Marks: 3****TOPIC: Compound Interest & Depreciation**

May 2024 P1HR - May 2024 | P1HR

- 5** Slavomir invests 5200 euros in a savings account for 4 years.
He gets 2.5% per year compound interest.

Work out how much money Slavomir will have in the savings account
at the end of 4 years.

Give your answer correct to the nearest euro.

..... euros

(Total for Question 5 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 5****Marks: 3****TOPIC: Compound Interest & Depreciation**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The multiplier for compound interest**

The multiplier for 2.5% compound interest is

$$1.025$$

2. Use compound interest

After 4 years,

$$5200(1.025)^4 = 5739.827 \dots$$

3. Use compound interest

Correct to the nearest euro,

$$5739.827 \dots \approx 5740$$

FINAL ANSWER

5740 euros

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Standard & Compound Units**

May 2024 P1HR - May 2024 | P1HR

- 6 The diagram shows a solid wooden cylinder.

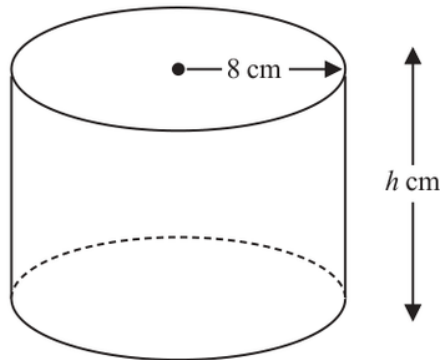


Diagram **NOT**
accurately drawn

The cylinder has radius 8 cm and height h cm.
The volume of the cylinder is 1208 cm^3

- (a) Work out the value of h
Give your answer correct to the nearest whole number.

$h = \dots\dots\dots$
(2)

The density of the wood is 1.25 g/cm^3

- (b) Work out the mass of the cylinder.
Give your answer in kilograms.

$\dots\dots\dots$ kilograms
(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 4****TOPIC: Standard & Compound Units**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

For the cylinder,

$$V = \pi r^2 h$$

$$1208 = \pi (8)^2 h$$

$$h = \frac{1208}{64\pi} = 6.006 \dots$$

2. Calculate value

So, to the nearest whole number,

$$h = 6$$

3. Calculate volume

For the mass, use the given volume:

$$\text{mass} = \text{density} \times \text{volume}$$

$$1.25 \times 1208 = 1510 \text{ g}$$

$$1510 \text{ g} = 1.51 \text{ kg}$$

FINAL ANSWER $h = 6 \text{ cm}$, and the mass is 1.51 kg .

PAST PAPER QUESTION**Q#: 7****Marks: 9****TOPIC: Factorising**

May 2024 P1HR - May 2024 | P1HR

7 (a) Simplify $g^9 \div g^2$
(1)(b) Expand $5k^2(k^3 + 4)$
(2)(c) (i) Factorise $x^2 - 2x - 63$
(2)(ii) Hence, solve $x^2 - 2x - 63 = 0$
(1)(d) Solve the inequality $7 - 2y < 3y - 12$
(3)**(Total for Question 7 is 9 marks)**

PAST PAPER SOLUTION**Q#: 7****Marks: 9****TOPIC: Factorising**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a) $g^9 \div g^2 = g^7$.

2. (b)

(b) $5k^2(k^3 + 4) = 5k^5 + 20k^2$.

3. (c)(i)

(c)(i)

$$x^2 - 2x - 63 = (x - 9)(x + 7)$$

4. Solve quadratic equation

(c)(ii) $x = 9$ or $x = -7$.

5. (d)

(d)

$$7 - 2y < 3y - 12$$

$$19 < 5y$$

FINAL ANSWER

(a) g^7 , (b) $5k^5 + 20k^2$, (c) $(x - 9)(x + 7)$, $x = 9$ or $x = -7$, (d) $y > \frac{19}{5}$

PAST PAPER QUESTION**Q#: 8****Marks: 6****TOPIC: Area & Perimeter**

May 2024 P1HR - May 2024 | P1HR

- 8 The diagram shows a trapezium, $ABCD$

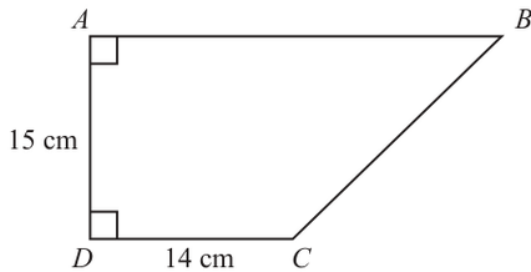


Diagram **NOT**
accurately drawn

DAB and ADC are right angles.

$$AD = 15 \text{ cm} \quad DC = 14 \text{ cm}$$

The area of the trapezium is 360 cm^2

Work out the perimeter of the trapezium.

..... cm

(Total for Question 8 is 6 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 6****TOPIC: Area & Perimeter**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use the trapezium area formula**

Use the trapezium area formula:

$$360 = \frac{1}{2}(AB + 14)(15)$$

$$AB + 14 = 48$$

$$AB = 34$$

2. Use trigonometryTriangle ABC is right-angled with horizontal length

$$34 - 14 = 20$$

3. Calculate area

So

$$BC = \sqrt{20^2 + 15^2} = 25$$

4. Perimeter

Perimeter:

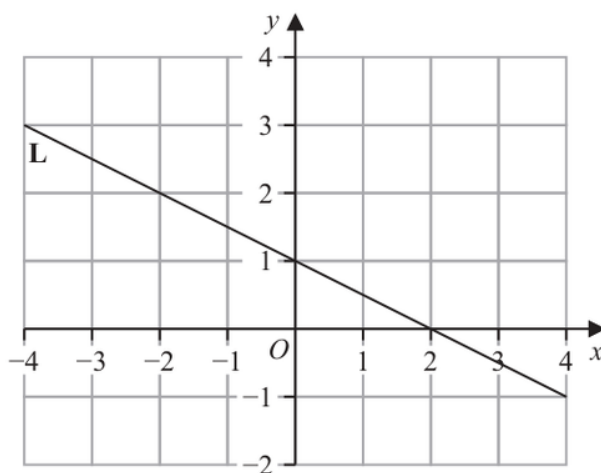
$$15 + 14 + 25 + 34 = 88$$

FINAL ANSWER

88 cm.

PAST PAPER QUESTION**Q#: 9** **Marks: 3****TOPIC: Linear Graphs ($y = mx + c$)**

May 2024 P1HR - May 2024 | P1HR

9 Line **L** is drawn on the grid.Find an equation for **L**Give your answer in the form $y = mx + c$

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Linear Graphs ($y = mx + c$)**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve equation**

From the graph, the line crosses the y-axis at 1, so

$$c = 1$$

2. It also crosses the x

It also crosses the x-axis at 2, so using (0, 1) and (2, 0),

$$m = \frac{0 - 1}{2 - 0} = -\frac{1}{2}$$

3. Solve equation

Therefore

$$y = -\frac{1}{2}x + 1$$

FINAL ANSWER

$$y = -\frac{1}{2}x + 1.$$

PAST PAPER QUESTION**Q#: 10****Marks: 2****TOPIC: Statistics Toolkit**

May 2024 P1HR - May 2024 | P1HR

10 Here are the numbers of goals scored by a hockey team in its 11 games this season.

0 1 2 2 3 4 4 6 7 9 11

Work out the interquartile range of the numbers of goals.

(Total for Question 10 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 10****Marks: 2****TOPIC: Statistics Toolkit**

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WORKED SOLUTION*Dr. Eslam Ahmed***1. The goals in order are**

The goals in order are

 $0, 1, 2, 2, 3, 4, 4, 6, 7, 9, 11$

$$Q_1 = 2, \quad Q_3 = 7$$

$$\text{IQR} = 7 - 2 = 5$$

FINAL ANSWER

5.

EA

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

May 2024 P1HR - May 2024 | P1HR

11 $A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

- (a) Write down the highest common factor (HCF) of $5A$ and $2B$
Give your answer as a product of prime factors.

.....
(2)

$A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

- (b) Work out the value of $(AB)^2$
Give your answer as a product of prime factors.

.....
(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 4****TOPIC: Prime Factors, HCF & LCM**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the LCM**

$$A = 2^5 \times 5 \times 7^2$$

$$B = 2^3 \times 5^3 \times 7^4$$

2. Find the LCM

For part (a),

$$5A = 2^5 \times 5^2 \times 7^2$$

$$2B = 2^4 \times 5^3 \times 7^4$$

3. The HCF takes the smaller

The HCF takes the smaller powers:

$$\text{HCF} = 2^4 \times 5^2 \times 7^2$$

4. Find the LCM

For part (b),

$$AB = 2^{5+3} \times 5^{1+3} \times 7^{2+4}$$

$$AB = 2^8 \times 5^4 \times 7^6$$

5. Find the LCM

Therefore

$$(AB)^2 = 2^{16} \times 5^8 \times 7^{12}$$

FINAL ANSWER(a) $2^4 \times 5^2 \times 7^2$, (b) $2^{16} \times 5^8 \times 7^{12}$

PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

May 2024 P1HR - May 2024 | P1HR

12 Solve the simultaneous equations

$$4x + 3y = 9.6$$

$$6x + 5y = 16.8$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

(Total for Question 12 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 12

Marks: 4

TOPIC: **Simultaneous Equations**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Solve simultaneous equations

$$4x + 3y = 9.6$$

$$6x + 5y = 16.8$$

2. Multiply the first equation by

Multiply the first equation by 5:

$$20x + 15y = 48$$

3. Multiply the second equation by

Multiply the second equation by 3:

$$18x + 15y = 50.4$$

4. Subtract

Subtract:

$$2x = -2.4$$

$$x = -1.2$$

5. Solve simultaneous equationsSubstitute into $4x + 3y = 9.6$:

$$4(-1.2) + 3y = 9.6$$

$$3y = 14.4$$

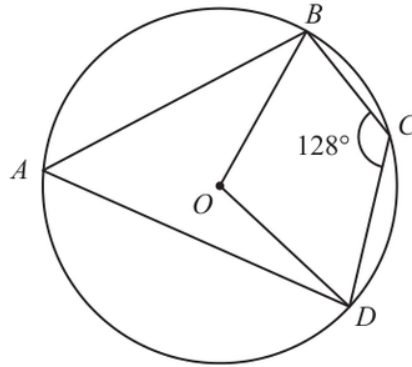
$$y = 4.8$$

FINAL ANSWER

$$x = -1.2, y = 4.8.$$

PAST PAPER QUESTION**Q#: 13****Marks: 5****TOPIC: Circle Theorems**

May 2024 P1HR - May 2024 | P1HR

13Diagram **NOT**
accurately drawn A , B , C and D are points on a circle, centre O Angle $BCD = 128^\circ$ Work out the size of angle OBD

Give a reason for each stage of your working.

angle $OBD = \dots\dots\dots^\circ$ **(Total for Question 13 is 5 marks)**

PAST PAPER SOLUTION**Q#: 13****Marks: 5****TOPIC: Circle Theorems**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Opposite angles in cyclic quadrilateral**Opposite angles in cyclic quadrilateral $ABCD$ add to 180° :

$$\angle BAD = 180^\circ - 128^\circ = 52^\circ$$

2. The angle at the centreThe angle at the centre is twice the angle at the circumference standing on the same arc BD :

$$\angle BOD = 2 \times 52^\circ = 104^\circ$$

3. Use trigonometrySince $OB = OD$, triangle OBD is isosceles:

$$\angle OBD = \frac{180^\circ - 104^\circ}{2} = 38^\circ$$

FINAL ANSWER 38° .

PAST PAPER QUESTION**Q#: 14****Marks: 6****TOPIC: Expanding Brackets**

May 2024 P1HR - May 2024 | P1HR

14 (a) Expand and simplify $(3x + 1)(2 - x)(4 + x)$
(3)(b) Simplify fully $\left(\frac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$
(3)

(Total for Question 14 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 14****Marks: 6****TOPIC: Expanding Brackets**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$(2 - x)(4 + x) = 8 - 2x - x^2$$

$$(3x + 1)(8 - 2x - x^2) = -3x^3 - 7x^2 + 22x + 8$$

2. (b)

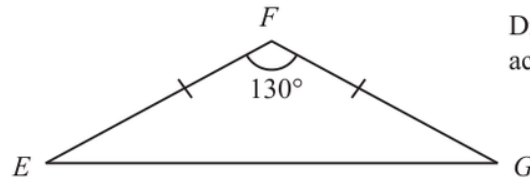
(b)

$$\left(\frac{a^3b}{a^9b^5}\right)^{-1/2} = \left(\frac{1}{a^6b^4}\right)^{-1/2} = a^3b^2$$

FINAL ANSWER(a) $-3x^3 - 7x^2 + 22x + 8$, (b) a^3b^2

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2024 P1HR - May 2024 | P1HR

15 The diagram shows isosceles triangle EFG Diagram **NOT**
accurately drawn

$$EF = GF$$

$$\text{Angle } EFG = 130^\circ$$

The area of triangle EFG is 74 cm^2 Work out the length of EF

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let**

Let

$$EF = FG = x$$

2. Use the area formula

Use the area formula:

$$74 = \frac{1}{2}x^2 \sin 130^\circ$$

$$x^2 = \frac{148}{\sin 130^\circ}$$

$$x = 13.899 \dots$$

FINAL ANSWER

13.9 cm.

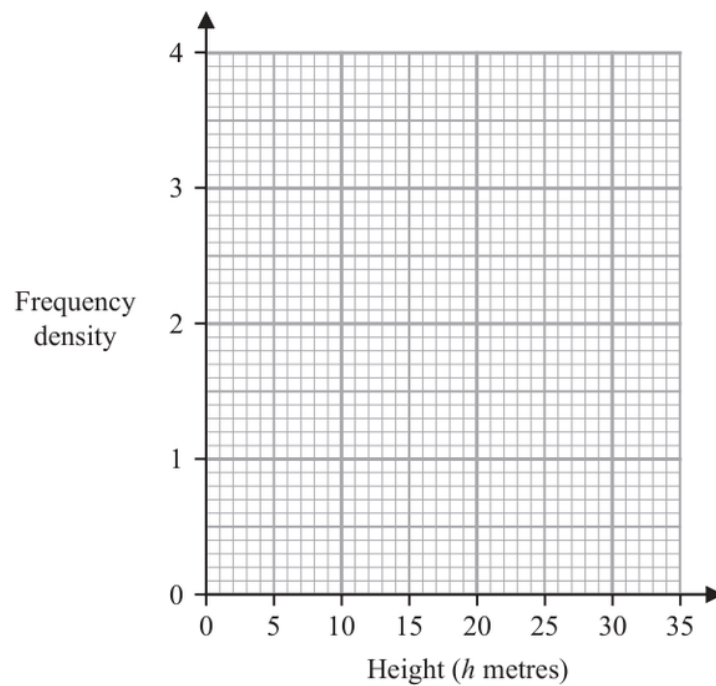
PAST PAPER QUESTION**Q#: 16****Marks: 3****TOPIC: Histograms**

May 2024 P1HR - May 2024 | P1HR

16 The table gives information about the heights, in metres, of the trees in a park.

Height (h metres)	Frequency
$0 < h \leq 2$	5
$2 < h \leq 5$	12
$5 < h \leq 10$	18
$10 < h \leq 20$	14
$20 < h \leq 35$	9

On the grid, draw a histogram for this information.



(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 3****TOPIC: Histograms**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use histogram**

For a histogram,

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

2. The bar heights are

The bar heights are:

$$0 < h \leq 2 : \frac{5}{2} = 2.5$$

$$2 < h \leq 5 : \frac{12}{3} = 4$$

$$5 < h \leq 10 : \frac{18}{5} = 3.6$$

$$10 < h \leq 20 : \frac{14}{10} = 1.4$$

$$20 < h \leq 35 : \frac{9}{15} = 0.6$$

FINAL ANSWER

draw bars with heights 2.5, 4, 3.6, 1.4, 0.6.

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Algebraic Roots & Indices**

May 2024 P1HR - May 2024 | P1HR

17 (a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of n

$$n = \dots\dots\dots (1)$$

(b) Express $\frac{7}{2-\sqrt{3}}$ in the form $\sqrt{c} + d$ where c and d are integers.

Show your working clearly.

$$\dots\dots\dots (3)$$

(Total for Question 17 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Algebraic Roots & Indices**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a) so**

(a)

$$\left(\sqrt[4]{k^{12}}\right)^5 = (k^3)^5 = k^{15}$$

so $n = 15$.**2. (b)**

(b)

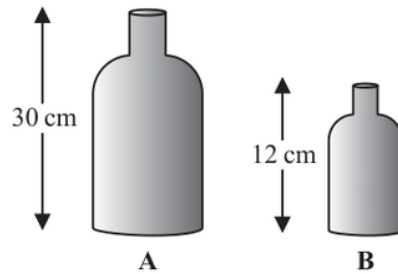
$$\frac{7}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}} = \frac{7(2 + \sqrt{3})}{4 - 3} = 14 + 7\sqrt{3}$$

$$7\sqrt{3} = \sqrt{49 \times 3} = \sqrt{147}$$

FINAL ANSWER(a) 15, (b) $\sqrt{147} + 14$

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

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18 The diagram shows two similar vases, **A** and **B**Diagram **NOT**
accurately drawnThe height of vase **A** is 30 cmThe height of vase **B** is 12 cm

Given that

$$\text{surface area of vase A} - \text{surface area of vase B} = 178.5 \text{ cm}^2$$

find the surface area of vase **A**..... cm²**(Total for Question 18 is 4 marks)**

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate area**

The linear scale factor from vase B to vase A is

$$\frac{30}{12} = 2.5$$

2. Calculate area

So the surface area scale factor is

$$2.5^2 = 6.25$$

3. Calculate surface area

Let the surface area of vase B be S .

$$6.25S - S = 178.5$$

$$5.25S = 178.5$$

$$S = 34$$

4. Calculate area

Therefore the surface area of vase A is

$$6.25 \times 34 = 212.5$$

FINAL ANSWER

212.5 cm².

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Differentiation**

May 2024 P1HR - May 2024 | P1HR

19 A curve C has equation $y = x^3 - 8x^2 - 12x + 5$

Curve C has exactly two stationary points, one at point A and one at point B such that

x coordinate of point $A > x$ coordinate of point B

Find the coordinates of point A

Show clear algebraic working.

(.....,)

(Total for Question 19 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 5****TOPIC: Differentiation**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Differentiate**

$$y = x^3 - 8x^2 - 12x + 5$$

$$\frac{dy}{dx} = 3x^2 - 16x - 12$$

2. Stationary points have

Stationary points have

$$3x^2 - 16x - 12 = 0$$

$$x = \frac{16 \pm \sqrt{(-16)^2 - 4(3)(-12)}}{6}$$

$$x = \frac{16 \pm 20}{6}$$

$$x = 6 \quad \text{or} \quad x = -\frac{2}{3}$$

3. DifferentiatePoint A has the greater x -coordinate, so $x = 6$.

$$y = 6^3 - 8(6^2) - 12(6) + 5$$

$$y = 216 - 288 - 72 + 5 = -139$$

FINAL ANSWER $A = (6, -139).$

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Completing the Square**

May 2024 P1HR - May 2024 | P1HR

- 20 (a)** Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a , b and c are numbers to be found.

.....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

- (b) Find the coordinates of P

(..... ,)

(2)

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 5****TOPIC: Completing the Square**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use Moved Coordinate Geometry Completing**

For part (a),

$$2x^2 - 11x + 9$$

2. Factor out 2 from the

Factor out 2 from the quadratic terms:

$$2\left(x^2 - \frac{11}{2}x\right) + 9$$

3. Complete the square

Complete the square:

$$2\left[\left(x - \frac{11}{4}\right)^2 - \frac{121}{16}\right] + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + \frac{72}{8}$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

4. Evaluate fraction

So

$$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

5. Calculate value

For part (b),

$$y = 2(x - 3)^2 - 11(x - 3) + 9$$

6. Let

Let

$$u = x - 3$$

7. Evaluate fraction

From part (a), the minimum occurs when

$$u = \frac{11}{4}$$

8. Evaluate fraction

So

$$x - 3 = \frac{11}{4}$$

$$x = \frac{23}{4}$$

9. The minimum value of is

The minimum value of y is

$$-\frac{49}{8}$$

FINAL ANSWER

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}, P = \left(\frac{23}{4}, -\frac{49}{8}\right)$$

EA

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Completing the Square**

May 2024 P1HR - May 2024 | P1HR

- 20** (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a , b and c are numbers to be found.

.....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

- (b) Find the coordinates of P

(..... ,)

(2)

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 5****TOPIC: Completing the Square**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use Moved Coordinate Geometry Completing**

For part (a),

$$2x^2 - 11x + 9$$

2. Factor out 2 from the

Factor out 2 from the quadratic terms:

$$2\left(x^2 - \frac{11}{2}x\right) + 9$$

3. Complete the square

Complete the square:

$$2\left[\left(x - \frac{11}{4}\right)^2 - \frac{121}{16}\right] + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + \frac{72}{8}$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

4. Evaluate fraction

So

$$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

5. Calculate value

For part (b),

$$y = 2(x - 3)^2 - 11(x - 3) + 9$$

6. Let

Let

$$u = x - 3$$

7. Evaluate fraction

From part (a), the minimum occurs when

$$u = \frac{11}{4}$$

8. Evaluate fraction

So

$$x - 3 = \frac{11}{4}$$

$$x = \frac{23}{4}$$

9. The minimum value of is

The minimum value of y is

$$-\frac{49}{8}$$

FINAL ANSWER

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}, P = \left(\frac{23}{4}, -\frac{49}{8}\right)$$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

21 There are 25 counters in a bag such that

6 counters are blue

x counters are orange, where $x > 9$

the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.

Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are 25 counters**

There are 25 counters.

2. Blue counters

Blue counters:

$$6$$

3. Orange counters

Orange counters:

$$x$$

4. Pink counters

Pink counters:

$$25 - 6 - x = 19 - x$$

5. The probability of taking one

The probability of taking one orange and one pink in either order is

$$\begin{aligned} \frac{x}{25} \cdot \frac{19-x}{24} + \frac{19-x}{25} \cdot \frac{x}{24} \\ = \frac{2x(19-x)}{25 \cdot 24} \end{aligned}$$

6. Evaluate fractionThis is equal to $\frac{22}{75}$, so

$$\frac{2x(19-x)}{600} = \frac{22}{75}$$

$$\frac{x(19-x)}{300} = \frac{22}{75}$$

$$x(19-x) = 88$$

$$19x - x^2 = 88$$

$$x^2 - 19x + 88 = 0$$

$$(x-8)(x-11) = 0$$

7. Solve inequality

Since $x > 9$,

$$x = 11$$

8. Calculate value

So the number of pink counters is

$$19 - 11 = 8$$

9. The probability of taking 2

The probability of taking 2 pink counters is

$$\begin{aligned}\frac{8}{25} \cdot \frac{7}{24} &= \frac{56}{600} \\ &= \frac{7}{75}\end{aligned}$$

FINAL ANSWER

$$\frac{7}{75}$$

EA

PAST PAPER QUESTION

Q#: 21

Marks: 5

TOPIC: **Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

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the rest of the counters are pink

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The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.

Show clear algebraic working.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Probability Toolkit**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are 25 counters**

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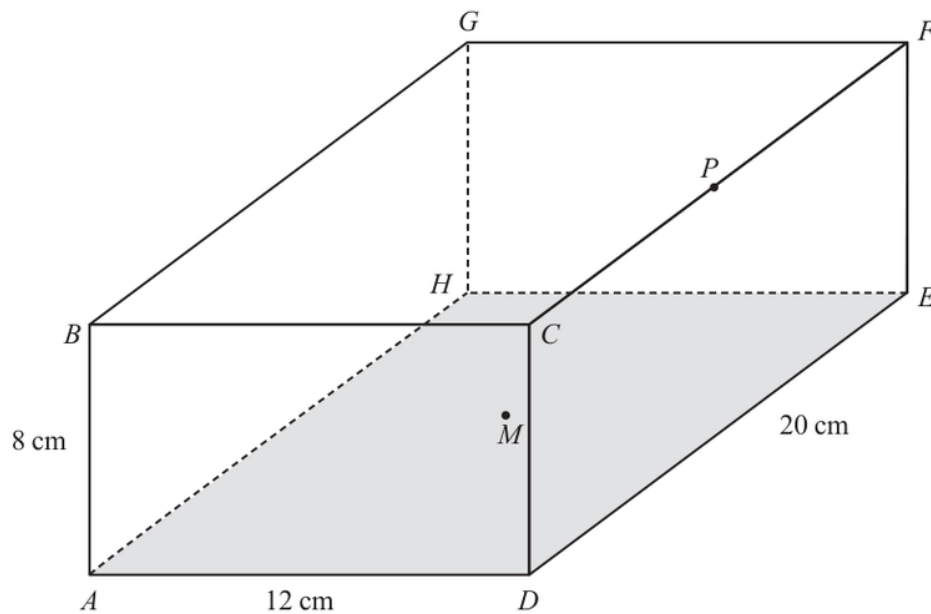
FINAL ANSWER

$$\frac{7}{75}$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: 3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

22 The diagram shows a cuboid $ABCDEFGH$ with horizontal base $ADEH$ Diagram **NOT**
accurately drawn

$AB = 8 \text{ cm}$

$AD = 12 \text{ cm}$

$DE = 20 \text{ cm}$

 M is the midpoint of the base $ADEH$ and P is the midpoint of the edge CF Work out the size of angle BMP

Give your answer correct to one decimal place.

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

Use coordinates with

$$A = (0, 0, 0)$$

2. Use trigonometry

Since

$$AD = 12, \quad DE = 20, \quad AB = 8$$

3. take

take

$$B = (0, 0, 8)$$

4. The centre of the rectangularThe centre of the rectangular base $ADEH$ is

$$M = (6, 10, 0)$$

5. Find the midpointPoint P is the midpoint of CF .

$$C = (12, 0, 8), \quad F = (12, 20, 8)$$

$$P = (12, 10, 8)$$

6. Use trigonometry

Now

$$\vec{MB} = B - M = (-6, -10, 8)$$

$$\vec{MP} = P - M = (6, 0, 8)$$

7. Use the scalar product formula

Use the scalar product formula:

$$\vec{MB} \cdot \vec{MP} = (-6)(6) + (-10)(0) + (8)(8)$$

$$\vec{MB} \cdot \vec{MP} = 28$$

$$|\vec{MB}| = \sqrt{(-6)^2 + (-10)^2 + 8^2} = \sqrt{200}$$

$$|\vec{MP}| = \sqrt{6^2 + 0^2 + 8^2} = 10$$

$$\cos \angle BMP = \frac{28}{10\sqrt{200}}$$

$$\angle BMP = 78.6^\circ$$

FINAL ANSWER 78.6°

EA

PAST PAPER QUESTION

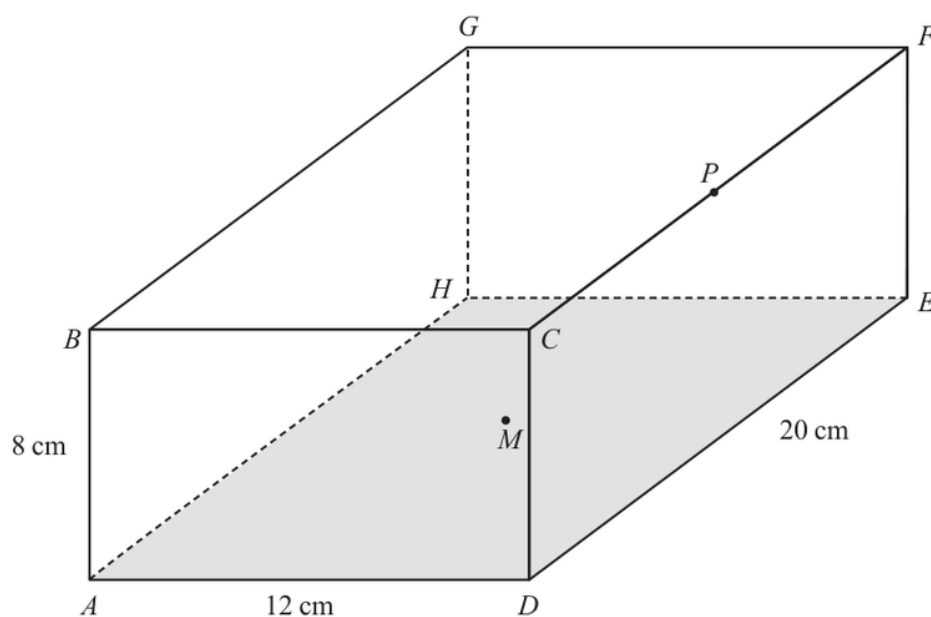
Q#: 22 Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

22 The diagram shows a cuboid $ABCDEFGH$ with horizontal base $ADEH$

Diagram **NOT**
accurately drawn



$AB = 8 \text{ cm}$ $AD = 12 \text{ cm}$ $DE = 20 \text{ cm}$

M is the midpoint of the base $ADEH$ and P is the midpoint of the edge CF

Work out the size of angle BMP

Give your answer correct to one decimal place.

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: 3D Pythagoras & Trigonometry**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

Use coordinates with

$$A = (0, 0, 0)$$

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4. The centre of the rectangularThe centre of the rectangular base $ADEH$ is

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5. Find the midpointPoint P is the midpoint of CF .

$$C = (12, 0, 8), \quad F = (12, 20, 8)$$

$$P = (12, 10, 8)$$

6. Use trigonometry

Now

$$\vec{MB} = B - M = (-6, -10, 8)$$

$$\vec{MP} = P - M = (6, 0, 8)$$

7. Use the scalar product formula

Use the scalar product formula:

$$\vec{MB} \cdot \vec{MP} = (-6)(6) + (-10)(0) + (8)(8)$$

$$\vec{MB} \cdot \vec{MP} = 28$$

$$|\vec{MB}| = \sqrt{(-6)^2 + (-10)^2 + 8^2} = \sqrt{200}$$

$$|\vec{MP}| = \sqrt{6^2 + 0^2 + 8^2} = 10$$

$$\cos \angle BMP = \frac{28}{10\sqrt{200}}$$

$$\angle BMP = 78.6^\circ$$

FINAL ANSWER **78.6°**

EA

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Sequences**

May 2024 P1HR - May 2024 | P1HR

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Sequences**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The first three terms are**

The first three terms are

$$4x - 14, \quad x + 2, \quad 7x - 9$$

2. Equal differences give

Equal differences give

$$(x + 2) - (4x - 14) = (7x - 9) - (x + 2)$$

$$-3x + 16 = 6x - 11$$

$$9x = 27$$

$$x = 3$$

3. Find the nth term

So the sequence is

$$-2, 5, 12, \dots$$

4. with common difference

with common difference 7.

$$S_{40} = \frac{40}{2}(2(-2) + 39(7))$$

$$S_{40} = 20(-4 + 273) = 5380$$

FINAL ANSWER

5380

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Sequences**

May 2024 P1HR - May 2024 | P1HR

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Sequences**

May 2024 P1HR - May 2024 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The first three terms are**

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FINAL ANSWER

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